

Liam Schoneveld

EDUCATION	Master of Science (Artificial Intelligence) <i>Cum Laude</i> The University of Amsterdam	2015-2017
	Bachelor of Commerce (Liberal Studies) The University of Sydney	2009-2013
PROFESSIONAL EXPERIENCE	Research Lead, <i>X-Reality</i> , Woven by Toyota	Jul 2022-Present
	- Own the technical direction and architecture of the avatar research program, defining research roadmaps and translating emerging methods into product capabilities. - Led research and development of a photorealistic single-image-to-3D-avatar system from research through production, supporting external-facing demonstrations and product development. - Led development of a novel real-time 3D head-tracking system from research through production, adopted internally and published as a first-author paper at ICCV 2025. - Serve as technical lead and mentor for a team of researchers and engineers, remaining hands-on in model architecture, implementation, experimentation, and technical decision-making. - Lead university research collaborations, aligning research direction with company goals and contributing to multiple top-tier publications.	
	Principal AI Researcher, Powder (\$14.5m 2021 Series A)	Oct 2019-Jul 2022
	- Led the AI division as a hands-on technical lead, driving research, model development, and technical direction across AI projects. - Built ML infrastructure for automated gaming highlight detection, including MLOps pipelines and a production labeling operation that remained central to Powder's product long after my departure. - Independently developed self-supervised learning approaches for automated video highlights detection, reducing model labeling requirements by 10x. - Led an academic collaboration with UPEC on audio-visual emotion recognition, resulting in two first-author publications.	
	Data Scientist, PandaScore (\$6m 2020 Series A)	Sep 2017-Oct 2019
	- Developed, deployed, and maintained computer vision systems for real-time event and statistics detection in e-sports tournaments. - Built real-time probabilistic inference systems using computer-vision-derived data to calculate and optimize betting odds for live matches.	
	Machine Learning Intern, Scyfer (acquired by Qualcomm)	Feb-Sep 2017
	Completed master's thesis <i>Semi-Supervised Learning With Generative Adversarial Networks</i> under the supervision of Taco Cohen and Prof. Max Welling.	
	Data Scientist, PricewaterhouseCoopers Australia	Mar 2013-Jan 2016
	Developed machine learning and analytics solutions across consulting engagements.	

PUBLICATIONS	Giebenhain, S., Kirschstein, T., Schoneveld, L. , Davoli, D., Chen, Z., & Nießner, N. <i>Pix2NPHM: Learning to Regress NPHM Reconstructions From a Single Image</i> . ECCV 2026.
	Schoneveld, L. , Chen, Z., Davoli, D., Tang, J., Terazawa, S., Nishino, K., & Nießner, M. (2025). <i>SHeaP: Self-Supervised Head Geometry Predictor Learned via 2D Gaussians</i> . ICCV 2025.
	Tang, J., Davoli, D., Kirschstein, T., Schoneveld, L. , & Nießner, M. (2025). <i>GAF: Gaussian Avatar Reconstruction From Monocular Videos via Multi-view Diffusion</i> . CVPR 2025.
	Qian, S., Kirschstein, T., Schoneveld, L. , Davoli, D., Giebenhain, S., & Nießner, M. (2024). <i>GaussianAvatars: Photorealistic Head Avatars With Rigged 3D Gaussians</i> . CVPR 2024.
	Schoneveld, L. , & Othmani, A. (2021). <i>Towards a General Deep Feature Extractor for Facial Expression Recognition</i> . IEEE ICIP 2021.
	Schoneveld, L. , Othmani, A., & Abdelkawy, H. (2021). <i>Leveraging Recent Advances in Deep Learning for Audio-Visual Emotion Recognition</i> . Pattern Recognition Letters, 146, 1-7.
TECHNICAL SKILLS	Schoneveld, L. (2017). <i>Semi-Supervised Learning With Generative Adversarial Networks</i> . Master's Thesis, The University of Amsterdam.
	Research: Deep Learning, Self-Supervised Learning, World Models, 3D Gaussian Splatting, Neural Rendering, 3D Reconstruction, 3D Human Modeling, Diffusion Models, NeRFs, Differentiable Rendering
	Engineering: Python, PyTorch, C++, CUDA, distributed training, real-time inference, model optimization, distillation, quantization
	ML Systems: NVIDIA Triton, TensorRT, ONNX, CoreML, Docker, MLflow, Weights & Biases, DVC, CI/CD
	Infrastructure: AWS, GCP, Azure, SageMaker, Terraform, Ray, NCCL